CONN Toolbox Implementation Methods

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User prompt: methods implemented in the CONN toolbox

Response: The **CONN toolbox** (widely used for resting-state and task-based functional magnetic resonance imaging analysis) implements an extensive, modular pipeline spanning spatial preprocessing, artifact denoising, first-level connectivity metrics, second-level statistical modeling, and cluster-level inferences.

The core methodological framework encompasses several key stages:

1. Spatial Preprocessing Pipelines

CONN provides standardized pipelines (compatible with SPM12, FSL, and external preprocessed outputs like fMRIPrep):

- **Realignment and Unwarping:** Corrects for subject head motion and susceptibility-by-movement interactions.
- **Slice-Timing Correction:** Adjusts for timing differences across slices acquired during the same TR.
- **Outlier Identification (Scrubbing):** Detects high-motion volumes using metrics like Framewise Displacement (FD) and Global Signal Change (DVARS) to flag outlier scans for regression or censoring.
- **Coregistration:** Aligns functional volumes to structural anatomical images.
- **Segmentation:** Classifies structural images into Gray Matter (GM), White Matter (WM), and Cerebrospinal Fluid (CSF) tissue classes.
- **Normalization:** Transforms data into standard MNI space (direct or indirect normalization via structural segmentations).
- **Smoothing:** Applies spatial Gaussian blurring kernels to functional data.

2. Denoising and Quality Control (The aCompCor Framework)

To minimize physiological noise, head motion artifacts, and non-neural fluctuations without relying on controversial Global Signal Regression (GSR), CONN implements:

- **Anatomical CompCor (aCompCor):** Extracts principal component time series from noise regions of interest (specifically, white matter and CSF masks) and regresses them out alongside their derivatives.
- **Motion Regression:** Regresses out estimated subject realignment parameters

(typically 6 motion parameters plus their first-order derivatives, or extended 24-parameter models).

- **Scrubbing/Censoring:** Includes artifact/outlier scrubbing covariates.
- **Band-Pass Filtering:** Restricts the frequency window of the BOLD signal (e.g., standard resting-state band of $0.01–0.1\ \mathrm{Hz}$).
- **Quality Assurance (QA) Metrics:** Generates BOLD carpet plots, FC-QC correlation metrics, global correlation (GCOR) measures, and framewise displacement distributions to evaluate denoising efficacy.

3. First-Level Functional Connectivity Measures

CONN calculates various individual-level connectivity indices, categorized by analysis type:

- **Seed-Based Connectivity (SBC):** Computes correlation or regression maps between a predefined seed region (ROI or voxel) and all other voxels in the brain.
- **ROI-to-ROI Connectivity:** Estimates connectivity matrices across sets of predefined anatomical or functional parcels.
- **Generalized Psycho-Physiological Interactions (gPPI):** Models task-specific or condition-dependent changes in functional connectivity.
- **Independent Component Analysis (ICA):** Performs group-level or individual-level spatial ICA (and masked ICA) to isolate intrinsic resting-state networks.
- **Voxel-to-Voxel Measures:**
    - **Intrinsic Connectivity Contrast (ICC):** Measures the overall connectivity of a voxel with the rest of the brain.
    - **Local Correlation (LC / ILC):** Evaluates local functional homogeneity.
    - **Amplitude of Low-Frequency Fluctuations (ALFF and fALFF):** Quantifies the intensity of regional low-frequency oscillations.
    - **Connectome-MVPA:** Uses multivariate pattern analysis to identify voxels with complex connectivity profiles across the brain.
- **Graph-Theoretical Analyses:** Automatically computes network properties such as global/local efficiency, clustering coefficient, path length, and modularity.

4. Second-Level General Linear Model (GLM) & Inferences

Group-level analyses utilize the General Linear Model to assess population-level effects:

- **Experimental Designs:** Supports regressions, two-sample t-tests, ANOVA, ANCOVA, and complex mixed within- and between-subject longitudinal designs.
- **Hypothesis Testing:** Evaluates linear contrasts of parameters (univariate and multivariate statistics).
- **Multiple Comparison Correction:** Implements family-wise error (FWE) control using both parametric methods (e.g., **Random Field Theory**) and robust non-parametric permutation/randomization techniques (including **Threshold-Free Cluster Enhancement [TFCE]**).

Sources

1. Learning the CONN Toolbox for Functional Connectivity Analysis
2. Handbook of functional connectivity Magnetic Resonance Imaging
3. (PDF) Conn : A Functional Connectivity Toolbox for Correlated and
4. fMRI Methods - CONN toolbox
5. CONN toolbox
